

- 2 (a) (i) 1. equal B1  
 2. density of ice is less than density of water B1

explanation: 1. at equilibrium, weight of ice = upthrust on ice

$$m_{\text{ice}}g = m_{\text{displaced water}}g$$

$$m_{\text{ice}} = m_{\text{displaced water}}$$

$$2. V_{\text{ice}}\rho_{\text{ice}} = V_{\text{displaced water}}\rho_{\text{water}}$$

since  $V_{\text{ice}} > V_{\text{displaced water}}$ , then  $\rho_{\text{ice}} < \rho_{\text{water}}$

- (ii) mass of ice becomes equal mass of water after melted/ B1  
 mass of melted ice equal mass of displaced water  
 volume of melted ice equals volume of water displaced/ B1  
 melted ice fills the space of water displaced by ice A1  
 so level does not change

- (b) (i) Upthrust is **equal in magnitude** and **opposite in direction** to the weight of the fluid displaced by a submerged or floating object. B1

- (ii) Upthrust = weight of water displaced by the anchor  
 = density of water x volume of anchor x  $g$   
 =  $1030 \times 0.50 \times 9.81$  C1  
 = 5050 N A1

- (iii) Let volume of air be  $V$   
 Upthrust on lifting bag + upthrust on anchor – weight of anchor =  $ma$   
 $(1030)V(9.81) + 5050 - 7800(0.50)(9.81) = 7800(0.50)(2.50)$  C1  
 $V = 4.25 \text{ m}^3$  A1

